

03: Inference for SLR

Stat 230 – Spring 2026

1 Part 1: On paper¹

Biologists know that the leaves on plants tend to get smaller as temperatures rise, and average temperatures have been getting hotter over time. A sample of 252 leaves from the species *Dodonaea viscosa* subspecies *angustissima* (broadleaf hopbush) have been collected in a certain region of South Australia. Is there an association between leaf width and year? Below is the coefficient table from a simple linear regression model.

term	estimate	std.error	statistic	p.value
(Intercept)	37.723	8.575	4.399	<0.001
Year	-0.018	0.004	-4.029	<0.001

Table 1: lm
output

Note: The width is the average width, in mm, of leaves, taken at their widest points, that were collected in a given year.

- Report the fitted regression equation.
- Interpret the slope of the regression line in context of the data.

¹Adapted from *Stat 2: Modeling with Regression and ANOVA*

- c. Calculate a 95% confidence interval for the slope of the regression line. (Use the slides to find the appropriate critical value). Interpret this interval in context of the data.
- d. What hypotheses are being tested with the statistic and associated p-value in the (Intercept) row of the table? What conclusion can you draw?